AI-Use Disclosure - Quantum Traffic Gridlock

1. AI Assistance Declared

I used Claude (Anthropic) via the claude.ai chat interface as an AI assistant during the hackathon. Event-specific formulation and code were developed during the event; no code, QUBO coefficients, schedules, or strategies from before the event were reused.

2. What the Assistant Produced

- **Initial Code & Logic:** Drafted the initial fluid-surrogate QUBO rule (effective-capacity cap, quadratic overflow proxy) and the first versions of `surr.py`, `run.py`, and the ablation scripts.
- **Initial Testing:** Ran the initial 5-seed lift/enrichment experiments in its sandbox and drafted the first version of the technical report.
- **Hyper-parameter Selection:** Suggested initial hyper-parameters ($k = 1.5$, $p = 3$, ramp initialization) based on early sandbox experiments.

3. Independent Verification and Refinement by the Team

- **Code Review:** I read and understood the main source files (`surr.py`, `run.py`, `baselines.py`, `check.py`) line-by-line to verify the affine queue evolution and Ising conversion logic.
- **Local Validation:** I independently executed `python baselines.py scenario_R.json` and `python check.py` on my local machine to verify the official scorer reference costs and the 0.00% surrogate regret claim.
- **Scenario H Execution:** I ran the unmodified pipeline on Scenario H to validate generalization and recorded the official results.
- **Editorial Control:** I edited the report to add explicit mathematical equations (q_t evolution, overflow proxy) to strengthen the derivation section and ensure all rubric criteria were clearly addressed.

4. Limits of AI Assistance

- All AI suggestions were treated as hypotheses and strictly verified against the official `scorer.py` and local simulators before inclusion.
- No hard-coded answers, seed lookups, or reading of scenario tables were used to decide outputs. All decisions derive strictly from the scenario-independent QUBO rules.
- Other tools used: Qiskit 2.x, Qiskit Aer, numpy, scipy, and standard Markdown-to-PDF exporters.